Mathematics practice booklet: Grade 7 expectations

Grade 7 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 7 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 7.EE.A.1, 7.EE.A.2, 7.EE.B.3, 7.EE.B.4

A worked example

Solve 3x + 5 = 20. Then check your answer.

1. Undo the + 5 by subtracting 5 from both sides: 3x = 15.
2. Undo the multiplication by dividing both sides by 3: x = 5.
3. Check: 3 × 5 + 5 = 15 + 5 = 20, which is correct.

x = 5

Questions

1. Simplify: 5x + 3 + 2x - 1

   A. 7x + 4

   B. 7x + 2

   C. 10x + 2

   D. 7x - 2

2. Expand: 4(x + 3)

   _____

3. Factor: 6x + 9

   A. 6(x + 3)

   B. 3(2x + 9)

   C. 9(x + 1)

   D. 3(2x + 3)

4. Expand and simplify: 2(3x - 4) + 5

   _____

5. Simplify: (5x - 2) - (2x - 6)

   _____

6. A taxi charges AED 7 to start the trip and AED 2 for each kilometre. Khalid's family pays AED 19 in all. How many kilometres was the trip?

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7. Solve: 3(x - 2) = 15

   A. x = 7

   B. x = 5

   C. x = 3

   D. x = 17

8. Solve: x / 4 - 3 = 5

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9. Solve the inequality: 2x + 3 > 11

   A. x > 7

   B. x < 4

   C. x > 4

   D. x > 14

10. Salem buys 3 cups of karak and one pastry that costs AED 4. He pays AED 19 in all. Write an equation and find the price of one cup.

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11. Maryam has AED 40 on her metro card. Each trip costs AED 3. What is the greatest whole number of trips she can take?

    A. 12

    B. 13

    C. 14

    D. 40

12. Hamdan says 3(x + 2) is the same as 3x + 2. Is he right? Explain.

Section 2: Shapes, measuring and space

Standards practised: 7.G.A.1, 7.G.A.2, 7.G.B.4, 7.G.B.5, 7.G.B.6

A worked example

A circular fountain has a radius of 7 m. Using pi = 22/7, find its circumference and its area.

1. Circumference = 2 x pi x r = 2 × 22/7 × 7 = 44 m.
2. Area = pi x r x r = 22/7 × 7 × 7 = 22 × 7 = 154.
3. Area is measured in square units: 154 square metres.

Circumference 44 m, area 154 square metres

Questions

1. On a map, 1 cm stands for 5 km. Two towns are 6 cm apart on the map. What is the real distance?

   A. 11 km

   B. 1.2 km

   C. 30 km

   D. 5 km

2. A room is 4 m by 3 m. In a scale drawing, 1 cm stands for 0.5 m. What are the sides of the room in the drawing?

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3. Which set of side lengths (in cm) can make a triangle?

   A. 3, 4, 8

   B. 2, 5, 9

   C. 6, 6, 13

   D. 5, 7, 10

4. A round pond in a park has a diameter of 10 m. Hamdan and his cousin walk once all the way round it. How far do they walk? Use pi = 3.14.

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5. A round coaster for a karak cup has a radius of 5 cm. What is its area? Use pi = 3.14.

   A. 31.4 square cm

   B. 78.5 square cm

   C. 314 square cm

   D. 15.7 square cm

6. A circular table has a circumference of 44 cm. Using pi = 22/7, what is its diameter?

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7. Two angles next to each other on a straight line add up to a straight angle. One of them is 115 degrees. What is the other?

   A. 65 degrees

   B. 55 degrees

   C. 75 degrees

   D. 245 degrees

8. Two straight lines cross. One of the angles is 50 degrees. Find the angle opposite it and the angle next to it.

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9. A floor is a rectangle 10 m by 6 m with a rectangular corner 4 m by 3 m cut out. What is the area of the floor?

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10. A triangular prism has a triangular end with base 6 cm and height 4 cm. The prism is 10 cm long. What is its volume?

    A. 240 cubic cm

    B. 24 cubic cm

    C. 60 cubic cm

    D. 120 cubic cm

11. Find the surface area of a box that is 5 cm by 4 cm by 3 cm.

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12. A water tank is 2 m long, 1.5 m wide and 1 m high. How many litres does it hold when full? (1 cubic metre = 1,000 litres)

Section 3: Data, graphs and chance

Standards practised: 7.SP.A.1, 7.SP.A.2, 7.SP.B.3, 7.SP.B.4, 7.SP.C.5, 7.SP.C.6, 7.SP.C.7, 7.SP.C.8

A worked example

A bag holds 2 red, 3 green and 5 yellow beads. One bead is taken without looking. What is the probability that it is green?

1. Count all the beads: 2 + 3 + 5 = 10.
2. Green beads: 3.
3. Probability = 3 out of 10 = 3/10 = 0.3.

3/10

Questions

1. A weather report says the probability of sunshine tomorrow is 0.9. What does this tell you?

   A. It is very likely to happen

   B. It is very unlikely to happen

   C. It is equally likely either way

   D. It is certain to happen

2. A bag has 4 red marbles and 6 blue marbles. One is picked without looking. What is the probability it is blue?

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3. A fair coin is tossed twice. What is the probability of two heads?

   A. 1/2

   B. 1/3

   C. 1/4

   D. 1/8

4. A fair coin is tossed once and a spinner with 3 equal colours (red, green, blue) is spun once. How many different outcomes are possible?

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5. Layla spins a spinner 50 times and it lands on green 18 times. What is the experimental probability of green?

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6. The probability of landing on green is 1/4. How many times would you expect green in 80 spins?

   A. 60

   B. 20

   C. 320

   D. 84

7. A school wants to know the favourite fruit of its 500 students. Which sample is better: 30 students chosen at random from the school list, or 30 students waiting at the fruit stall? Why?

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8. In a random sample of 40 students, 10 prefer football. How many of the 600 students in the school would you predict prefer football?

   A. 60

   B. 240

   C. 400

   D. 150

9. Team A has a mean height of 150 cm and Team B has a mean height of 156 cm. Both groups have a mean absolute deviation of 4 cm. The difference between the means is how many times the mean absolute deviation?

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10. Group A scores: 5, 6, 7, 8, 9. Group B scores: 8, 9, 10, 11, 12. How much higher is the median of group B?

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11. A seed sprouts with a probability of 40%. You use random digits from 0 to 9 to simulate one seed. Which choice of digits stands for sprouting?

    A. Only the digit 4

    B. The digits 0 to 4

    C. The digits 0 to 3

    D. The digits 4 to 9

12. Hamdan tosses a fair coin and gets heads 4 times in a row. He says tails is now more likely. Is he right? Explain.

Section 4: Numbers and calculating

Standards practised: 7.RP.A.1, 7.RP.A.2, 7.RP.A.3, 7.NS.A.1, 7.NS.A.2, 7.NS.A.3

A worked example

A bag costs AED 80 before VAT. VAT is 5%. What is the price with VAT?

1. VAT = 5% of 80 = 0.05 × 80 = AED 4.
2. Price with VAT = 80 + 4 = AED 84.
3. Another way: multiply by 1.05: 80 × 1.05 = 84.

AED 84

Questions

1. A boy walks 1/2 km in 1/4 of an hour. What is his speed in km per hour?

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2. 3 kg of dates cost AED 54 at the souq. At the same price, what do 5 kg cost?

   A. AED 90

   B. AED 72

   C. AED 270

   D. AED 108

3. A table pairs x and y: (2, 6), (4, 12), (5, 16). Is y proportional to x? Explain.

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4. A shirt costs AED 120. It is sold at 25% off. What is the sale price?

   A. AED 90

   B. AED 30

   C. AED 145

   D. AED 95

5. A pair of shoes costs AED 240 before VAT. What is the total with 5% VAT?

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6. A trader at the souq buys a rug for AED 40 and marks it up by 30%. What is the selling price?

   A. AED 12

   B. AED 52

   C. AED 70

   D. AED 28

7. Noura saves AED 2,000 at 4% simple interest a year. How much interest does she earn in 3 years?

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8. The price of a ticket rises from AED 50 to AED 60. What is the percent increase?

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9. A diver is 12 m below the sea surface (-12 m). She swims up 5 m. Where is she now?

   A. -17 m

   B. -12 m

   C. -7 m

   D. 7 m

10. Work out: (a) -4 - (-9) (b) (-6) x (-7)

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11. Work out: -3/4 + 1/2

    A. -5/4

    B. 1/4

    C. -2/6

    D. -1/4

12. A jacket costs AED 150. It is reduced by 20%, and then 5% VAT is added to the reduced price. What does Omar pay? Show the steps.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. B. 7x + 2. Add the x terms: 5x + 2x = 7x. Add the numbers: 3 - 1 = 2. If your child chose A (7x + 4), they added 3 + 1 and ignored the minus.
2. Multiply 4 by each term: 4 x x = 4x and 4 × 3 = 12, so 4x + 12.
3. D. 3(2x + 3) 3 divides both 6 and 9, so 6x + 9 = 3(2x + 3). If your child chose B (3(2x + 9)), they did not divide the 9 by 3.
4. Expand: 2 × 3x = 6x and 2 x (-4) = -8, so 6x - 8. Then add 5: 6x - 8 + 5 = 6x - 3.
5. Subtracting a bracket flips each sign inside: 5x - 2 - 2x + 6. Then 5x - 2x = 3x and -2 + 6 = 4, so 3x + 4.
6. 2k + 7 = 19. Subtract 7: 2k = 12. Divide by 2: k = 6, so the trip was 6 km.
7. A. x = 7. Divide both sides by 3: x - 2 = 5, then add 2: x = 7. If your child chose B (x = 5), they forgot to add the 2 back.
8. Add 3 to both sides: x / 4 = 8. Multiply both sides by 4: x = 32.
9. C. x > 4. Subtract 3 to get 2x > 8, then divide by 2: x > 4. If your child chose B (x < 4), they flipped the sign, which is only needed when dividing by a negative.
10. 3c + 4 = 19. Subtract 4: 3c = 15. Divide by 3: c = 5, so a cup costs AED 5.
11. B. 13. 3n ≤ 40 gives n ≤ 13.3, and trips are whole, so 13 trips cost 39. If your child chose C (14), they rounded 13.3 up, but 14 trips cost 42, more than she has.
12. No. The 3 multiplies both terms: 3 x x = 3x and 3 × 2 = 6, so 3(x + 2) = 3x + 6. Try x = 1: 3(3) = 9 but 3x + 2 = 5.

Section 2: Shapes, measuring and space

1. C. 30 km. Real distance is map distance times the scale: 6 × 5 = 30 km. If your child chose A (11 km), they added 6 + 5 instead of multiplying.
2. Each 0.5 m becomes 1 cm, so divide by 0.5: 4 / 0.5 = 8 cm and 3 / 0.5 = 6 cm.
3. D. 5, 7, 10. The two shorter sides must add to more than the longest: 5 + 7 = 12, which is more than 10. If your child chose A (3, 4, 8), they did not test it: 3 + 4 = 7 is less than 8, so the sides cannot meet.
4. The distance round a circle is its circumference: pi x diameter = 3.14 × 10 = 31.4 m.
5. B. 78.5 square cm. Area = 3.14 × 5 × 5 = 78.5 square cm, using the radius. If your child chose C (314), they used the diameter 10 as the radius.
6. Circumference = pi x diameter, so diameter = 44 / (22/7) = 44 × 7/22 = 14 cm.
7. A. 65 degrees. Angles on a straight line add to 180, so 180 - 115 = 65 degrees. If your child chose D (245), they used 360 instead of 180.
8. Opposite angles are equal, so it is 50 degrees. Angles on a line add to 180, so the next one is 180 - 50 = 130 degrees.
9. Whole rectangle: 10 × 6 = 60. Cut out piece: 4 × 3 = 12. Subtract: 60 - 12 = 48 square metres.
10. D. 120 cubic cm. End area = 1/2 × 6 × 4 = 12, and volume = 12 × 10 = 120 cubic cm. If your child chose A (240), they forgot the half in the triangle.
11. Three pairs of faces: 5 × 4 = 20, 5 × 3 = 15 and 4 × 3 = 12. Double the total: 2 × 47 = 94 square cm.

12. Volume = 2 × 1.5 × 1 = 3 cubic metres. Each cubic metre is 1,000 litres, so 3 × 1,000 = 3,000 litres.

Section 3: Data, graphs and chance

1. A. It is very likely to happen. 0.9 is close to 1, so sunshine is very likely, but not certain. If your child chose D (certain), they treated 0.9 as 1.
2. There are 4 + 6 = 10 marbles and 6 are blue, so the chance is 6/10, which simplifies to 3/5.
3. C. 1/4. The four outcomes HH, HT, TH, TT are equally likely and only HH has two heads, so 1/4. If your child chose A (1/2), they looked at one toss only.
4. Multiply the choices: 2 outcomes for the coin times 3 for the spinner, 2 × 3 = 6.
5. Experimental probability is hits over tries: 18/50. Dividing both by 2 gives 9/25, which is 0.36.
6. B. 20. The expected number is 1/4 of 80, which is 80 / 4 = 20. If your child chose A (60), they found 3/4 of 80.
7. The random list. Every student has a fair chance, so it should look like the whole school, while students at the fruit stall may like fruit more than others.
8. D. 150. 10/40 = 1/4 of the sample, so 1/4 of 600 = 150. If your child chose A (60), they multiplied 10 × 6 without using the sample size of 40.
9. The difference between the means is 156 - 150 = 6. Divide by the MAD: 6 / 4 = 1.5 times.
10. The median is the middle score: 7 for group A and 10 for group B. 10 - 7 = 3.
11. C. The digits 0 to 3. 40% is 4 out of 10, so 4 digits stand for sprouting: 0, 1, 2, 3. If your child chose B (0 to 4), they counted 5 digits, which is 50%.
12. No. A coin has no memory, so each toss is independent. The chance of tails is still 1/2 on the next toss.

Section 4: Numbers and calculating

1. Speed is distance divided by time: 1/2 / 1/4 = 1/2 × 4 = 2 km per hour.
2. A. AED 90. Find 1 kg first: 54 / 3 = 18, then 5 kg cost 18 × 5 = 90. If your child chose C (270), they multiplied 54 by 5 and forgot it was the price of 3 kg.
3. For a proportion, y / x must always be the same. 6/2 = 3 and 12/4 = 3 but 16/5 = 3.2, so it is not proportional.
4. A. AED 90. 25% of 120 is 30, and the sale price is 120 - 30 = 90. If your child chose B (30), they stopped at the discount.
5. VAT is 5% of 240 = 0.05 × 240 = 12. Total: 240 + 12 = AED 252.
6. B. AED 52. 30% of 40 is 12, and a markup is added: 40 + 12 = 52. If your child chose A (12), they gave only the extra amount, not the selling price.
7. Interest each year is 4% of 2,000 = 80. For 3 years: 80 × 3 = AED 240.
8. The increase is 60 - 50 = 10. Compare it with the starting price: 10 / 50 = 0.2, which is 20%.
9. C. -7 m. Going up 5 from -12 gives -12 + 5 = -7, which is 7 m below the surface. If your child chose A (-17 m), they went down instead of up.
10. (a) Subtracting -9 is adding 9: -4 + 9 = 5. (b) Two negatives multiply to a positive: 6 × 7 = 42.
11. D. -1/4. Use quarters: 1/2 = 2/4, so -3/4 + 2/4 = -1/4. If your child chose A (-5/4), they subtracted 1/2 instead of adding it.
12. 20% off: 150 × 0.8 = 120. Then 5% VAT on 120: 120 × 1.05 = 126, so Omar pays AED 126.